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A Systematic Review and Meta-Analysis of Wearable Satellite System Technology for Linear Sprint Profiling: Technological Innovations and Practical Applications

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Abstract

Cormier, P, Meylan, C, Agar-Newman, D, Geneau, D, Epp-Stobbe, A, Lenetsky, S, and Klimstra, M. A systematic review and meta-analysis of wearable satellite system technology for linear sprint profiling: technological innovations and practical applications. *J Strength Cond Res* XX(X): 000–000, 2023—An emerging and promising practice is the use of global navigation satellite system (GNSS) technology to profile team-sports athletes in training and competition. Therefore, the purpose of this narrative systematic review with meta-analysis was to evaluate the literature regarding satellite system sensor usage for sprint modeling and to consolidate the findings to evaluate its validity and reliability. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, an electronic search of the databases, PubMed and SPORTDiscus (EBSCO), was conducted. Concurrent validity and reliability studies were considered, and 16 studies were retained for the review from the initial 1,485 studies identified. The effects on outcomes were expressed as standardized mean differences (SMDs, Cohen's *d*) for each outcome (i.e., maximal sprint speed [MSS], the acceleration constant [τ], maximal theoretical velocity [V_0], relative force [F_0], and relative power [P_{max}]). Effect magnitudes represented the SMD between GNSS-derived and criterion-derived (i.e., radar and laser) and resulted in the following estimates: small for MSS (d = 0.22, 95% CI 0.01 to 0.42), τ (d = −0.18, 95% CI −0.60 to 0.23), V_0 (d = 0.14, 95% CI −0.08 to 0.36), relative F_0 (d = 0.15, 95% CI −0.25 to 0.55), and relative P_{max} (d = 0.21, 95% CI −0.16 to 0.58). No publication bias was identified in meta-analyzed studies and moderator analysis revealed that several factors (sampling rate and sensor manufacturer) influenced the results. Heterogeneity between studies was considered moderate to high. This highlighted the differences between studies in sensor technology differences (i.e., sampling rate, sensor fusion, and satellite network acquisition), processing techniques, criterion technology used, sprint protocols, outcome reporting, and athlete characteristics. These findings may be useful in guiding improvements in sprint modeling using GNSS technology and enable more direct comparisons in future research. Implementation of all-out linear sprint efforts with GNSS technology can be integrated into sport-specific sessions for sprint modeling when robust and consistent data processing protocols are performed, which has important implications for fatigue monitoring, program design, systematic testing, and rehabilitation in individual and team sports.

Key Words: athlete monitoring, running, biomechanics, sensors, rehabilitation

Introduction

The integration of wearable microsensor technology has become commonplace within elite sports teams to monitor and quantify athlete movement and inform sport science and medicine disciplines (4,17,23,40,43,53). Global navigation satellite systems (GNSSs) and Global Positioning Systems (GPSs), collectively referred to as GNSS for the purposes of this article, are some of the most commonly used tracking sensors in sports. These sensors provide measurements of athlete acceleration, velocity, and displacement which are often the basis for training and competition load monitoring metrics (4,24). Alongside training and match metrics, GNSS sensors can be used to assess sprint acceleration capabilities. Accordingly, all-out sprint protocols are commonly performed, from which measures of distance, time, peak velocity, and acceleration are quantified (26,43). An example of which is horizontal force-velocity (FV) profiling, an approach that models athlete speed or distance over time from an all-out sprint allowing for the estimation of an athlete's theoretical horizontal maximal force and velocity capability (52). Although FV profiling is becoming a well-established technique, various field and laboratory technologies (e.g., radar, timing gates, treadmill, and force plates) have been preferred to collect force, speed, or timing data to model with the FV approach (26,52). Because GNSS technology

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is commonplace in daily training environments and is a technology capable of measuring athlete acceleration and velocity (22,39), there is great potential for the use of this tool to measure sprints and model GNSS velocity data using the FV approach. However, there is uncertainty as to the validity and reliability of GNSS technology regarding sprint modeling due to the lack of consistency in its application with no reviews consolidating the findings from available studies thus far.

Although practitioners have been applying sprint modeling techniques using GNSS data in the field, and wearable sensor manufacturers have begun to incorporate sprint profiling into their user outputs, there is a need to determine the current state of GNSS technology for FV profiling (5,11,13,36). To confidently use GNSS technology for sprint modeling, an assessment of the validity and reliability of this technology must be performed because the metrics outputted may be affected by the technical specifications of the devices, sprint protocols, and data-processing approaches (37). As there are many GNSS sensors available from manufacturers, studies continually validate sensors using different methodologies and protocols with various cohorts of individual and team-sports athletes (15,16). In addition, although there has been a focus on the validation and potential application of GNSS technology for FV profiling, there are inconsistent findings. For example, early attempts to construct sprint profiles with GNSS have been subject to measurement error affecting the validity and reliability of systems because of low sampling rates (e.g., ≤5 Hz) and limited satellite network capacity (37). As GNSS technology evolves, with increased sampling rates (≥10 Hz) and multiple sensor usage (e.g., fusion of inertial measurement units [IMUs] with GNSS) to improve accuracy and precision, the potential for its effective use for the purposes of sprint modeling is improving.

Given the discrepancies in study methodology, changes in this technology over recent years, and lack of recommendations on sprint modeling protocols using GNSS. The purpose of this review was to systematically assess the current literature available, exploring satellite positioning system sensor usage for sprint modeling, and to consolidate and summarize the findings as well as identify gaps in the literature to guide future research. Therefore, a systematic review and meta-analysis with narrative review was performed to provide a comprehensive and contextualized understanding of this subject for researchers and practitioners in the field.

Methods

Experimental Approach to the Problem

A systematic-narrative hybrid review with meta-analysis approach was taken where a comprehensive systematic search linked to the research question was performed. Articles were selected based on uniform criteria, each selected critically and evaluated with elements of qualitative and quantitative synthesis with evidence-based inferences (55). Furthermore, moderating analyses were performed to determine whether methodological factors affected the effect estimates. The ethics committee of the University of Victoria declared that approval was not necessary.

Data Sources and Search Strategy

This review was guided by a systematic search of the literature following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (46). An electronic search of the databases, PubMed and SPORTDiscus (EBSCO), was conducted including articles published up until July 1st, 2023, using the following Boolean search phrase: ("*GPS*" *OR* "*GNSS*" *OR* "*global positioning system*" *OR* "*global navigation satellite system*" *OR* "*accelerometer*" *OR* "*inertial measurement unit*" *OR* "*sensor fusion*") *AND* ("*sprinting*" *OR* "*team-sports*" *OR* "*athletes*"). A search of reference lists from relevant literature was also performed to identify articles that fit the criteria.

Inclusion and Exclusion Criteria

Inclusion criteria included were (a) usage of GNSS or GPS system for maximal linear sprint acceleration assessment, (b) validity and reliability estimates reported, (b) studies published in English, (c) over ground sprint mechanical characteristics (e.g., horizontal acceleration, force, velocity, or power) derived from continuous velocity-time GNSS data reported as an outcome measure, and (d) peer-reviewed literature. Exclusion criteria were the following: (a) subjects had any musculoskeletal disorders, (b) caliber of subjects fell below the classification of "recreationally active" as defined by recently proposed frameworks (38), (c) other positioning systems such as local positioning systems or optical tracking systems were used, and (d) surveys or reviews. This protocol was registered with the Open Science Framework (osf.io/tsg6r).

Study Selection

Databases were searched by 1 author (P.C.). After removal of duplicates, abstracts and titles were screened independently by 2 authors (P.C. and D.G.) based on the inclusion and exclusion criteria. A third party settled any disagreements on study inclusions (M.K.). After this initial screen, all article pdfs were sourced and evaluated by 2 authors (P.C. and D.G.) and data extracted by 1 author (P.C.).

Study Quality Assessment

The quality of the included studies was assessed with a modified Downs and Black (18) checklist, similar to previous validity study quality assessment reviews (15,33). The items 1, 2, 3, 6, 7, 10, 16, 18, and 20 were selected. Items were scored as 1 yes, 0 no, and 0 unable to determine. Item 20 was only applied if the study assessed validity. The study bias and quality were assessed by 2 authors (P.C. and D.G.) with any disagreements resolved by a third author (M.K.). Description of items can be found in Supplemental Digital Content 1 (see Table 1, http://links.lww.com/JSCR/A453).

Data Extraction

The FV metrics extracted were the maximal sprint speed (MSS), acceleration constant (τ), theoretical maximal horizontal force at zero velocity (F_0), theoretical maximum velocity at zero force (V_0), theoretical maximal power (P_{max}), and decrease in ratio of force (Drf %). These were reported in absolute or normalized to body mass. Refer to the work by Morin and Samozino (43) for more details as well as further explanations and interpretations of FV metrics. Studies that included peak velocity estimates from all-out linear sprint validity and reliability studies outside of the FV profiling approach were also included and reported as MSS, so long as the data were derived from continuously collected instantaneous GNSS data (i.e., not obtained from speed "zones").

The data extracted regarding the agreement of measurement was mean bias from Bland-Altman plots with confidence intervals (CIs) with limits of agreement (LOA). Bias was computed from

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mean differences between averages from the field measure being assessed against a criterion measure. Intraclass correlation coefficient (ICC), typical error of measurement (TEM), and typical error (TE) (sometimes expressed as the coefficient of variation [CV]) were also extracted from studies. For the sake of clarity, the reliability (precision) was termed "intersystem" when comparing between manufacture units, "intrasystem" when assessing reliability between units from the same manufacturer, "intertrial" when determining between-trial reliability, and "intermodel" when assessing different models from the same manufacturer. Attempts were made to contact study authors in cases where information was missing relevant to the review.

Statistical Analysis

Meta-Analysis. Analyses were conducted in R statistical software (version 4.2.2; R Core Team) (48). Only studies using valid FV profiling criterions capable of accurately measuring speed (i.e., radar and laser) (25) were eligible for the analysis. Outcomes were only included if more than 1 study reported the metrics; therefore, Drf, absolute F_0, and P_{max} were left out of the analysis. Retained data were analyzed using a mixed-effects meta-analysis (*metafor* package (58)) to determine the overall difference of effect on outcomes (i.e., MSS, τ, V_0, relative F_0, and relative P_{max}) between the criterion and the GNSS-derived FV metrics. The effect sizes were quantified and expressed as standardized mean differences (SMDs, Cohen's *d*) with 95% CIs. The weighted average effect size estimate was calculated, and a significant z-statistic test ($p < 0.05$) determined whether the summary estimate was significantly different from zero, where a non significant difference indicated overall agreement between the outcomes (GNSS) compared with the criterion. Threshold values for effect sizes (SMD) were 0.2–0.49 (small), 0.5–0.79 (medium), and ≥0.8 (large) (12). A moderator analysis was then performed to determine whether sampling rate (≥10 Hz vs. <10 Hz) or sensor manufacturer affected the outcomes of the model (62).

Heterogeneity (I^2) between studies and between study variance (τ^2) was evaluated. Heterogeneity was interpreted on a scale of magnitude of low <25%, medium 25–75%, and high >75% (29,30). A *p* value of <0.1 for the χ^2 indicated the presence of heterogeneity and a $\tau^2 > 1$ suggests substantial heterogeneity (30). Publication bias was evaluated through an asymmetry test as estimated from a funnel plot and egger test (19), a $p < 0.05$ was considered statistically significant.

Narrative Analysis. Using the entire data set, a narrative and quantitative summary was used to identify trends in the agreement (validity) and reliability studies. This approach was chosen because of the heterogeneity in types of reported outcomes for validity and reliability. As a priori, we interpreted the extracted percent agreements as good <5%, moderate 5–10%, and poor >10% based on thresholds from previous GNSS validation research (7,8,13). Extracted FV profile metrics' mean and standard deviations for the criterion and the GNSS measurement are displayed if reported in the included studies (Table 2, Figures 2–6).

To determine whether changes in performance were meaningful, some studies in this review have reported TE/CV (i.e., noise) and group mean and standard deviation results to calculate the smallest worthwhile change (SWC), calculated as the between-subject *SD*·0.2, and have further assessed the signal-to-noise ratio of the collected metric data (e.g., SWC/TE) (51) (Table 4). This was extracted from the studies if reported with signal-to-noise calculated or if TE and SWC were both reported. Signal-to-noise ratio was classified as <0.80 = poor, 0.80–1.00 = okay, and >1.00 = good and CVs <5% considered good (47,51). Intraclass correlation coefficients were classified as <0.50 = poor, 0.50–0.75 = moderate, 0.75–0.90 = good, and >0.90 = excellent (34).

Results

Study Inclusion

The search strategy yielded 1,458 results after removal of duplicates. After initial removal of articles by title and abstract inspection, based on the exclusion criteria, only studies assessing linear sprinting with GNSS were retained ($n = 42$). After this initial screen, all article pdfs were sourced and evaluated. After this final screen, 16 studies were retained for the review (Figure 1).

Table 1
Summary of subject characteristics and study quality.*

Reference	Subjects (*n*)	*f* (%)	Caliber	Sport	Age (y)	Height (m)	Body mass (kg)	Study quality (%)
Fonasier-Santos et al (21)	18	28	Trained to elite	Sprinters	27 ± 6.7	1.79 ± 0.1	72.0 ± 12.6	89
Cormier et al (13)	55	55	Trained to highly trained/National (University)	Rugby, ice Hockey	20.4 ± 2.4 and 22.3 ± 1.6	1.68 ± 0.06 and 1.83 ± 0.07	74.8 ± 12.0 and 87.2 ± 9.6	100
Cormier et al (14)	15	100	Elite/International	Soccer	26.23 ± 5.19	1.68 ± 0.07	63.1 ± 4.8	100
Clavel et al (11)	16	0	Elite/International	Rugby	NI	1.78 ± 0.08	78.3 ± 13.2	100
Apre et al (3)	9	22	Trained to elite	Sprinters	NI	NI	NI	100
Alphin et al (2)	24	100	Highly trained (university)	Lacrosse	19.7 ± 1.2	1.66 ± 0.06	64.8 ± 6.5	100
Huggins et al (32)	15	0	Highly trained (university)	Soccer	20.0 ± 1.0	1.77 ± 0.08	71.6 ± 7.2	100
Beato et al (8)	10	0	Recreationally trained	Team sports	22.0 ± 1.0	1.75 ± 0.06	71.8 ± 5.0	89
Barr et al (5)	28	0	Highly trained (university)	Football	19.0–24.0	NI	81.0–141.0	100
Lacome et al (36)	15	NI	Elite/International	Rugby 7s	26.0 ± 5.0	1.81 ± 0.08	90.0 ± 12.0	100
Reinhardt et al (49)	34	0	Highly trained	Soccer	20.0 ± 4.0	1.80 ± 0.06	74.0 ± 7.0	89
Hoppe et al (31)	6	0	Trained	Team sports	27.0 ± 2.0	1.77 ± 0.04	80.0 ± 2.8	89
Nagahara et al (45)	32	0	Trained to highly trained	Rugby, soccer	20.0 ± 2.2	1.76 ± 0.07	73.7 ± 9.7	89
Beato (6)	10	0	Trained	Soccer	24.0 ± 1.5	1.84 ± 0.06	80.0 ± 8.6	88
Akenhead et al (1)	1	0	Highly trained	Soccer	21.0	1.79	85.0	89
Waldron et al (60)	19	0	Highly trained	Rugby league	14.7 ± 0.5	1.88 ± 0.07	72.8 ± 10.7	100

**f* (%) = percentage of sample that was women; NI = no information available.

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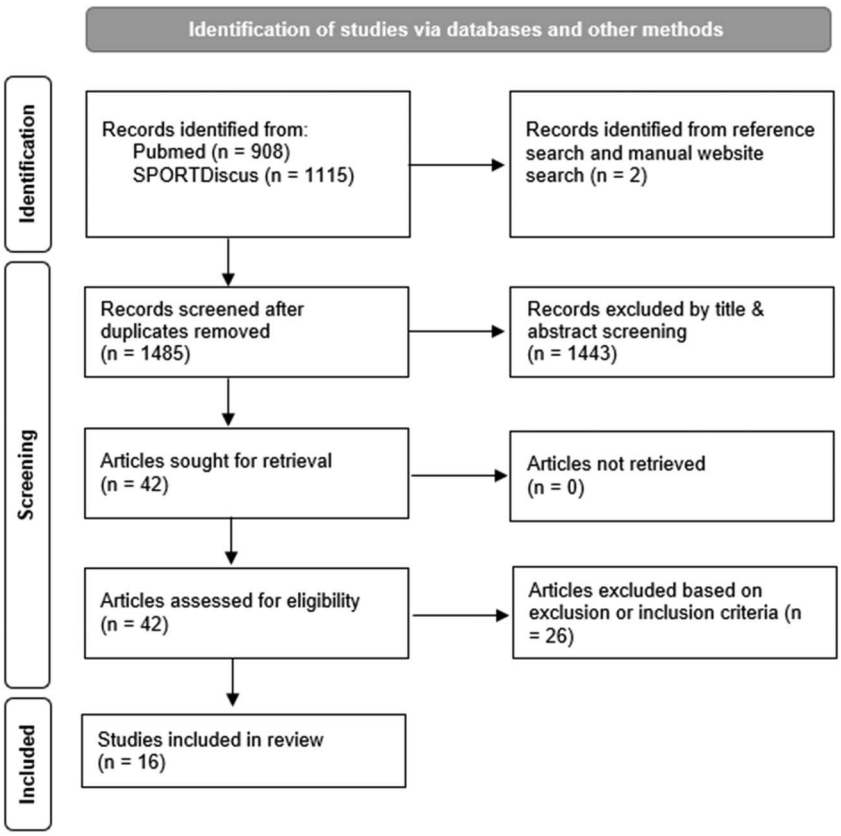

Figure 1. Study selection flowchart (PRISMA 2020). PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Subject and Study Characteristics

Of the 16 studies included, all studies were concurrent validity or reliability studies assessing the accuracy of a GNSS field system to a criterion measurement device or assessing various types of reliability. Namely, the criteria were a radar gun (3,11,13,36,45), laser (32,45,56), video kinematic analysis (6), sliding platform (1), and timing gates (2,3,5,31,48). Two studies used IMUs to correct or improve their GNSS velocity and acceleration estimates (3,49). Trial numbers ranged from 2 to 3, with subjects performing 20–100-m sprints depending on the aims of the study. Trials were performed on different surfaces such as rubberized tracks (13), artificial grass (1,31,36), grass (11,35,45), or unspecified track surfaces (3,5,6,8). All protocols were performed outdoors.

Quality Assessment

Overall, all identified studies were deemed of good quality (89–100%); therefore, no studies were omitted after the inclusion and exclusion criteria omissions (Table 1). Criteria from items 18 (appropriate statistical test reporting) and 20 (appropriate criterion) were most often violated. Scoring details are provided in Supplemental Digital Content 1 (see Table 1, http://links.lww.com/JSCR/A453).

Meta-Analysis and Moderator Analysis

Pooled effect estimates, representing differences between GNSS-derived values compared with criterion-derived, effect magnitudes were small for MSS (d = 0.22, 95% CI 0.01 to 0.42, I^2 = 55%), τ (d = −0.18, 95% CI −0.60 to 0.23, I^2 = 87%), V_0 (d = 0.14, 95% CI −0.08 to 0.36, I^2 = 56%), F_0 (d = 0.15, 95% CI −0.25 to 0.55, I^2 = 87%), and P_{max} (d = 0.21, 95% CI −0.16 to 0.58, I^2 = 80%) (Figures 2–6). It must be noted that heterogeneity was considered high for τ, F_0, and P_{max} and moderate for V_0 and MSS. Visual inspection of the funnel plots and egger tests did not suggest publication bias.

Moderator analysis revealed a moderate overestimation of MSS for SensorEverywhere (d = 0.59, 95% CI 0.18 to 0.99) and SPI (d = 0.75, 95% CI 0.38 to 1.11) with a moderate overestimation in systems with sampling rates <10 (d = 0.75, 95% CI 0.38 to 1.11) (Table 5). τ was overestimated moderately with StatSports (d = 0.68, 95% CI 0.32 to 1.04) and largely underestimated using GPEXE (d = −0.86, 95% CI −1.18 to −0.53). V_0 was overestimated with SPI (d = 0.73, 95% CI 0.37 to 1.10) with a moderate overestimation in systems with sampling rates <10 (d = 0.73, 95% CI 0.37 to 1.10). F_0 was largely overestimated with GPEXE (d = 0.95, 95% CI 0.62 to 1.27) and moderately underestimated by StatSports (d = −0.64, 95% CI −0.99 to −0.28). P_{max} was moderately overestimated with GPEXE (d = 0.69, 95% CI 0.38 to 1.01.

Narrative Assessment

All outcomes (including all possible criteria) were extracted and reported in Table 3 and percent agreement displayed in Figure 7. It was found that the highest bias for acceleration-based outcomes was demonstrated when timing gates were used as a criterion. System sensors sampling at 5 Hz showed larger bias than sensors

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Table 2
FV profiling metric comparison between field measure and criterion for each validity study.*†

Reference (Hz)	τ (s)		V_0 (m·s^{-1})		F_0 (N)		F_0 (N·kg^{-1})		P_{max} (W)		P_{max} (W·kg^{-1})		Drf (%)	
	Criterion	Field	Criterion	Field	Criterion	Field	Criterion	Field	Criterion	Field	Criterion	Field	Criterion	Field
Fornasier-Santos et al. (21) (10 Hz)	1.24 (0.12)	1.29 (0.16)	8.30 (1.24)	8.33 (1.28)	—	—	6.40 (0.78)	6.18 (0.96)	—	—	—	—	—	—
Cormier et al. (13) (10 Hz)	1.50 (0.2)	1.51 (0.27)	8.24 (0.87)	8.17 (0.89)	429.3 (88.6)	427.3 (94.7)	5.21 (0.63)‡	5.18 (0.77)‡	886.5 (216.7)	875.5 (230.7)	10.77 (1.96)‡	10.62 (2.20)‡	−6.05 (0.81)	−6.06 (0.95)
Clavel et al. (11) (10 Hz)	1.24 (1.14)	1.25 (1.08)	9.26 (1.06)	9.23 (1.06)	—	—	6.98 (1.11)	6.96 (1.07)	—	—	16.7 (1.12)	16.08 (1.10)	—	—
Hoppe et al. (31) (10 Hz)	1.1 (0.0)	1.38	8.2 (0.1)	8.2	—	—	7.7 (0.1)	6.42	—	—	16.1 (0.4)	13.48	—	—
Hoppe et al. (31) (18 Hz)	1.1 (0.0)	1.38	8.2 (0.1)	8.04	—	—	7.7 (0.1)	6.39	—	—	16.1 (0.4)	13.38	—	—
Nagahara et al. (45) (20 Hz)	1.29 (0.10)	1.41 (0.17)	8.92 (0.44)	8.96 (0.56)	—	—	6.63 (0.51)	6.10 (0.60)	—	—	14.68 (1.41)	13.66 (1.51)	—	—
Nagahara et al. (45) (5 Hz)	1.13 (0.08)	1.10 (0.30)	8.93 (0.43)	8.46 (0.79)	—	—	7.63 (0.41)	7.96 (2.17)	—	—	17.02 (1.23)	16.69 (4.13)	—	—
Cormier et al. (13) (10 Hz)	1.50 (0.2)	1.36 (0.21)	8.24 (0.87)	8.06 (0.82)	429.3 (88.6)	471.5 (113.9)	5.21 (0.63)‡	5.71 (0.91)‡	886.5 (216.7)	953.3 (262.2)	10.77 (1.96)‡	11.55 (2.48)‡	−6.05 (0.81)	−6.70 (1.01)

*FV = force-velocity
†Data in mean (*SD*) or solely mean because *SD* not reported or only % difference reported.
‡Unpublished results.
Criterion = radar, Field = GNSS or GPS

sampling at 10 Hz and above when compared to radar or laser. Two studies used sensor fusion which resulted in a minimal mean bias of −0.01 m·s^{-1} for MSS, −0.17 m·s^{-1} for V_0, −0.02 N·kg^{-1} for relative F_0, and −0.31 W·kg^{-1} for relative P_{max} compared with radar or timing gates (3,49). Filtering or smoothing the GNSS velocity signal resulted in better estimates than the use of raw data (1,36). Modeling metrics such as measures of peak velocity or speed were considered in good agreement, and measures of acceleration or force were considered in moderate agreement compared with a criterion measure (e.g., radar/laser). Poor percent agreement was observed for acceleration, force, and power metrics when timing gates or a laser was used as the criterion (Figure 6).

Regarding intertrial reliability outcomes, altogether, studies reported CVs <5% for MSS, V_0, and relative P_{max}, whereas F_0 and τ were >5% (Table 4). Intrasystem CV comparing raw vs. smoothed velocity 10 Hz signal was 15.6 and 3.1%, respectively. 16 Hz raw and smoothed velocity resulted in a CV of 0.5% for both raw and smooth (filtered) velocity (ICC = 0.99 for both). Intraclass correlation coefficients demonstrated *good* to *excellent* reliability; however, acceleration-based metrics (i.e., τ, F_0, and Drf) displayed *good* reliability. One study assessed intermodel reliability for MSS which resulted in a CV of 0.15% and an ICC of 0.98 (8). The signal-to-noise ratio was considered "good" for MSS, V0, and Pmax, but "poor" for F0 and t (5,11,36) (Tables 4 Figures 7 and 8).

Discussion

This review evaluated the state of wearable satellite technology for athlete sprint modeling applications. There are currently 16 studies assessing aspects of validity and reliability of different GNSS sensors for sprint modeling. Quantitative analysis revealed that pooled effects showed good agreement between GNSS-derived outcomes with criterion measurements. Moderator analysis further revealed influences of sampling rates (≥10 Hz vs <10 Hz) and sensor manufacturers. Common evaluated outcomes point to agreement of modeling metrics such that measures of peak velocity or speed are considered in good agreement, and measures of acceleration and force are considered in moderate agreement compared with a criterion measure (e.g., radar and laser). Most studies shared good intertrial, intermodel, and intrasystem reliability. However, the CVs reported show potential inconsistency either within athlete variation or random error rooted within the technology itself. Although the consolidated studies in this review provide normative values and ranges for sprint modeling using GNSS, there are major identified differences that highlight gaps and opportunities for improvement of this technology. Important differences between studies relate to sensor technology differences (e.g., sampling rate, sensor fusion, and satellite network acquisition), processing techniques, criterion technology used, sprint protocols, outcome reporting, and athlete characteristics (37,57). These identified considerations may be useful to guide the improvement of the state of this technology and enable more direct comparison of results.

Based on the consolidated outcomes in this review, it has been shown that small SMDs were found for all meta-analyzed outcomes. When considering the entire data set, there are favorable results regarding GNSS criterion agreement for maximum velocity (MSS and V_0), whereas acceleration-derived metrics, such as τ, F_0, P_{max}, and Drf, achieve good to moderate agreement for percentage differences. These results may be closely tied to the

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Table 3

Concurrent validity of each metric using absolute (A) and/or mean differences (%) between field measurements with criterion by system.*†

Reference	System (filter)	Hz	Criterion (filter)	MSS ($m \cdot s^{-1}$)		τ (s)		V_0 ($m \cdot s^{-1}$)		F_0 (N)		F_0 ($N \cdot kg^{-1}$)		P_{max} (W)		P_{max} ($W \cdot kg^{-1}$)		Drf (%)	
				A	%	A	%	A	%	A	%	A	%	A	%	A	%	A	%
Fornasier-Santos et al. (21)	Catapult (S)	10	Radar/Laser (NI/S)	−0.01/ 0.03§	0.97/1.06‡	0.05/ 0.13	9.51/ 12.89‡	0.03/ 0.06	1.27/ 1.18‡			−0.21/− 0.59	8.63/ 11.54‡						
Cormier et al. (13)	Catapult (S)	10	Radar (4o0B)	−0.06§	−0.84§	0.01	0.38	−0.07	−0.89	−2	−0.96	−0.03¶	−0.96¶	−11.1	−1.85	−0.14¶	−1.85¶	−0.01	−0.27
Clavel et al. (11)	Catapult (S)	10	Radar (4o0B)	−0.05§	−0.28§	0.04	0.51	−0.04	−0.27			−0.02	−0.19			−0.05	−0.56		
Hoppe et al. (31)	Catapult (*1HzcoB*)	10	TG (NA)				24.7		0				−16.6				−16.3		
Akenhead et al. (1)	Catapult (R, S)	10	SP (NA)		0.2_{raw}, 0.03_{smooth}														
Apte et al. (3)	FieldwizII (KM)	10	Radar, TG (KM, NA)					−0.17				−0.02				−0.31			
Hoppe et al. (31)	GPEXE (*1HzcoB*)	18	TG (NA)				25.4		−0.2				−17				−16.9		
Nagahara et al. (45)	GPEXE (NI)	20	Radar, Laser (NI)	0.01#	0.2§	0.12#	9.7	0.04#	0.5			−0.53	−7.9			−1.02	−7.9		
Huggins et al. (32)	Polar Team Pro (R)	10	TG & Laser (NA, NI)	0.1															
Reinhardt et al. (49)	Polar Team ProII (K)	10	TG	−0.01															
Lacome et al. (36)	SensorEverywhere (R, MA)	16	Radar (MA)		-3.0_{raw}, -1.6_{smooth}														
Barr et al. (5)	SPI (NI)	5	TG (NA)	0.09															
Waldron et al. (60)	SPI (NI)	5	TG (NA)	0.56															
Nagahara et al. (45)	SPI Pro (NI)	5	Radar, Laser (NI)	−0.42	−4.8§	−0.03	−3.2	−0.47	−5.1			0.33	4.1			−0.33	−2.1		
Cormier et al. (13)	STATSports (R)	10	Radar (4o0B)	−0.13§	1.62§	−0.14	−9.81	−0.18	−2.18	42.23	8.62	0.5¶	8.62¶	66.82	6.46	0.79¶	6.46¶	−0.65	9.96
Beato et al. (6)	STATSports (R)	10	Video (NA)		3 to 9														
Alphin et al. (2)	VX Sport (MS)	10	TG (NA)		0.09														

*GPS = Global Positioning System; GNSS = global navigation satellite system; IMU = inertial measurement units; S = smooth manufacturer filter; R = raw data; SP = sliding platform; TG = timing gate; NI = no information, 4o0B = digital fourth-order zero lag Butterworth; 1HzcoB = digital low-pass two-passes 1 Hz cutoff Butterworth; NA = not applicable; = Kalman; M = Madgwick; MA = 0.5 s moving average; MS = software from manufacturer.

†Sorted by GPS or GNSS and sampling rate.

‡Difference with laser, raw velocity export, smooth-specific filtering, or smoothing technique used.

§Estimated.

||Reported sensor fusion approach with IMUs.

¶Unpublished results.

#Calculated after the fact.

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Reference	System	Hz	Criterion M	Criterion SD	GNSS M	GNSS SD	n	Weight	Estimate [95% CI]
Nagahara et al., (45).2	SPI	5	8.54	0.4	8.12	0.68	62	12.7%	0.75 [0.38, 1.11]
Fornasier-Santos et al., (21).1	Catapult	10	8	1.15	8.02	1.18	36	10.3%	-0.02 [-0.48, 0.44]
Fornasier-Santos et al., (21).2	Catapult	10	7.99	1.15	8.02	1.18	36	10.3%	-0.03 [-0.49, 0.44]
Cormier et al., (13).1	Catapult	10	7.84	0.84	7.78	0.79	63	13.2%	0.07 [-0.28, 0.42]
Cormier et al., (13).2	StatSports	10	7.84	0.84	7.72	0.74	63	13.1%	0.15 [-0.20, 0.50]
Clavel et al., (11)	Catapult	10	8.84	1.06	8.81	1.05	32	9.7%	0.03 [-0.46, 0.52]
Lacome et al., (36).1	SensorEverywhere	16	8.37	0.26	8.11	0.39	25	8.1%	0.77 [0.20, 1.35]
Lacome et al., (36).2	SensorEverywhere	16	8.22	0.23	8.09	0.38	25	8.3%	0.41 [-0.15, 0.97]
Nagahara et al., (45).1	GPEXE	20	8.61	0.39	8.62	0.42	80	14.3%	-0.02 [-0.33, 0.29]
Heterogeneity: $Tau^2 = 0.05$; $Chi^2 = 17.88$, df = 8 (P = 0.02); $I^2 = 55\%$									0.22 [0.01, 0.42]
Test for overall effect: Z = 2.03 (P = 0.04)									

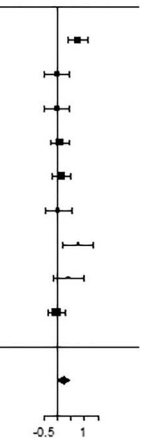

Figure 2. Forest plot with summary effect for MSS. MSS = maximal sprint speed.

current limitations of the sensor technology. For example, sampling rate is an important factor in sprint profiling with GNSS sensors. Nagahara et al. (45) identified that lower sampling rates (<10 Hz) resulted in poor agreement, with some profiling metrics with sampling rates ≥10 Hz providing superior agreement and reliability. More recent studies have since shown improvements in agreement outcomes (CVs <5%) across metrics with improved intrasystem and intersystem reliability using 10 Hz GNSSs (11,13,21). Although measures of speed are achieving good agreement with higher sampling rates, present results suggest that GNSS is still suboptimal for quantifying acceleration metrics (20,57). It has been suggested that low sampling rates limit the ability to determine changes in velocity during the initial portions of the sprint where acceleration is likely the highest. As acceleration is the rate of change in velocity, limitations in adequate sample rate will have implications for calculation of accurate acceleration. This can be observed in technology such as timing gates where low temporal resolution during the initial portion of the sprint will tend to overestimate F_0 in sprint modeling studies (13). As shown in Figure 3, both intertrial and intrasystem values for all metrics improve with increased sampling rates. These results suggest that optimal sampling rates for GNSS (and other sensors)-derived sprint modeling still need to be determined. As current systems range between 10 and 20 Hz, systems with greater sampling rates may need to be developed or enhanced.

Most GNSSs include IMUs which consist of an accelerometer, gyroscope, and magnetometer. Inertial measurement units can provide an accurate and useful measurement of sensor orientation and motion within a short distance and time range and are considered assistive to GNSS technology. However, IMUs, as opposed to GNSS technology, are not yet suitable to determine absolute changes in distance and speed over large ranges and each sensor technology can play useful and complementary roles in athlete monitoring (17). Recently, the outputs of GNSS and IMU sensor systems have been combined ("sensor fusion") and used to correct the GNSS signal output, increase the sampling rate, and provide additional kinematic information for output variables. For example, Apte et al. (3) and Reinhardt et al. (49) both used a sensor fusion approach for signal correction in their study with a 200 Hz IMU and 10 Hz GNSS unit using Madgwick and Kalman filtering which resulted in very high agreement between the metrics derived from the GNSS and the criterion selected. Although more research is warranted to implement sensor fusion for sprint modeling, its application could resolve many sprint modeling limitations of GNSS technology.

Reference	System	Hz	Criterion M	Criterion SD	GNSS M	GNSS SD	n	Weight	Estimate [95% CI]
Nagahara et al., (45).2	SPI	5	1.13	0.08	1.1	0.3	62	14.8%	0.14 [-0.22, 0.49]
Fornasier-Santos et al., (21).1	Catapult	10	1.24	0.12	1.29	0.16	36	13.7%	-0.35 [-0.82, 0.12]
Fornasier-Santos et al., (21).2	Catapult	10	1.17	0.11	1.29	0.16	36	13.5%	-0.86 [-1.35, -0.38]
Cormier et al., (13).1	Catapult	10	1.5	0.2	1.51	0.27	63	14.8%	-0.04 [-0.39, 0.31]
Cormier et al., (13).2	StatSports	10	1.5	0.2	1.36	0.21	63	14.7%	0.68 [0.32, 1.04]
Clavel et al., (11)	Catapult	10	1.24	1.14	1.25	1.08	32	13.4%	-0.01 [-0.50, 0.48]
Nagahara et al., (45).1	GPEXE	20	1.29	0.1	1.41	0.17	80	15.0%	-0.86 [-1.18, -0.53]
Heterogeneity: $Tau^2 = 0.27$; $Chi^2 = 51.04$, df = 6 (P = 0.00); $I^2 = 87\%$									-0.18 [-0.60, 0.23]
Test for overall effect: Z = -0.85 (P = 0.39)									

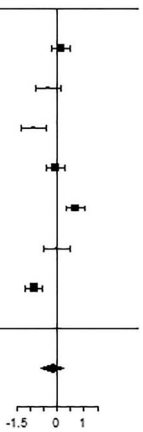

Figure 3. Forest plot with summary effect for τ.

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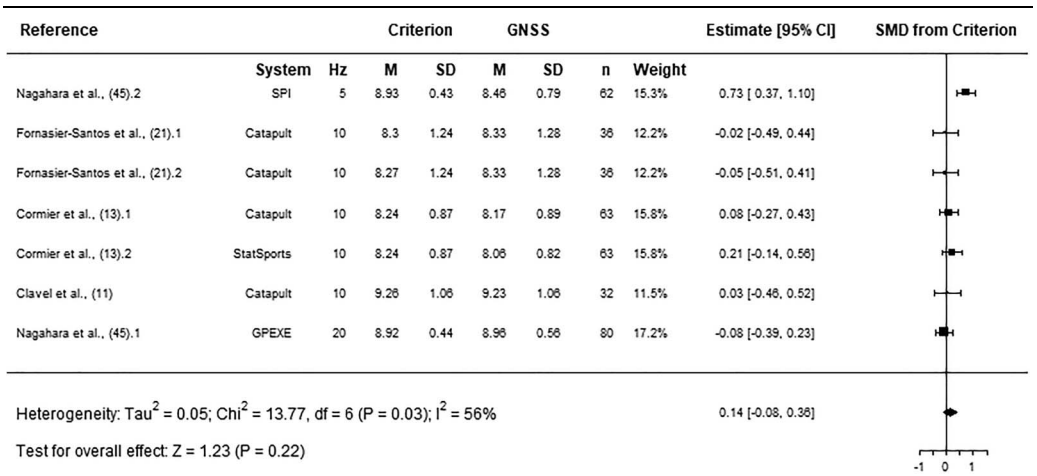

Figure 4. Forest plot with summary effect for V_0.

There are many considerations about the GNSS data used and how data are processed before sprint modeling outputs are derived. First, it is important to consider what kinematic variables are used as sprint modeling input parameters. Current sprint modeling practices require continuous acceleration or velocity, or distance or time data. Acceleration values exported from GNSS manufacturer software may be derived from either embedded accelerometers or velocity and time samples (61). Velocity is estimated using 2 methods, positional differentiation (distance/time) or Doppler shift, with the latter preferred because of superior accuracy (54). It is important to differentiate between these 2 methods because some studies report them interchangeably or concurrently which should influence the methodology, interpretation of findings, and study comparisons. Furthermore, acceleration and velocity outputs from manufacturer software are subject to proprietary filtering and algorithms as well as post-processing techniques used by researchers, as such the specific processing and cleaning methods may be unknown and may change with different software updates (37).

Another aspect that could affect sprint modeling outcomes is data filtering (37). Most GNSS manufacturers use undisclosed or proprietary filtering techniques, resulting in differently processed signals from sensors. Therefore, some sensors may output raw GNSS calculations while others have onboard preprocessing before the user is given the data. These preprocessing steps are often undisclosed, potentially hindering postexport applications conducted by sport scientists. Although the level of preprocessing is usually unknown, it is still common practice for sport scientists to use signal post-processing to further condition the data for sprint modeling. There were a variety of approaches identified in this review with unclear methodology being reported. Of note, Butterworth filters (31), rolling averages (e.g., 0.5 seconds) (36), and locally weighted regression smoothing (loess) (11) were used in the signal processing protocols. Otherwise, manufacturers' "smooth" export or raw data was used (Table 3). Studies in this review demonstrate that filtered data improve agreement with criterion measurement. For example, Lacome et al. (36) and Akenhead et al. (1) observed that MSS and maximum acceleration biases were larger with raw data vs. smoothed (moving average) data during linear sprinting tasks. In accordance with previous literature (20), a low-pass Butterworth filter was the most frequently reported filtering technique with both radar and GNSS data in this review. However, as mentioned, and important to note, only 2 GNSS studies reported all specific filtering specifications (e.g., digital, low-pass, dual-pass, or 1 Hz cutoff Butterworth; Table 3). Other studies exported the raw data or "smoothed" data from the manufacturers. In addition, regarding the

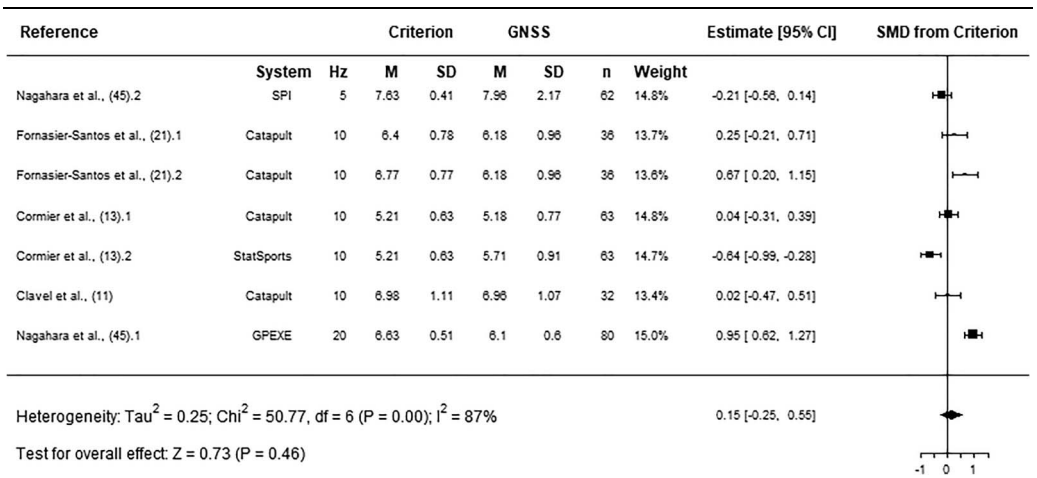

Figure 5. Forest plot with summary effect for relative F_0.

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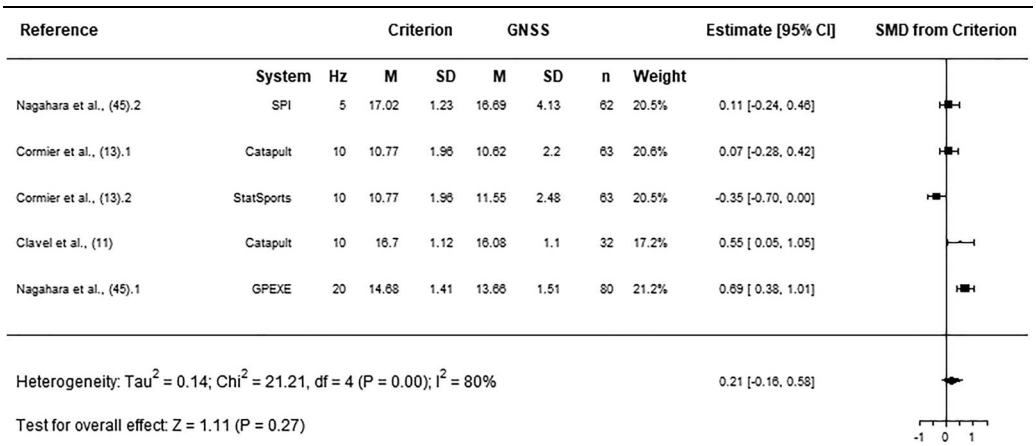

Figure 6. Forest plot with summary effect for relative P_{max}.

criterion systems (e.g., radar), important information, such as cutoff frequency, filter order, and number of passes, was also frequently left out, which makes comparison of study findings difficult. Similar to sampling rate considerations, the choice of filter and filter frequency is of utmost importance in ensuring that the true signal is not attenuated, and accurate sprint modeling metrics can be developed

Table 4

Reported reliability estimates from each study.*

Refs	System, Hz	Reliability type	MSS (m·s^{-1})	τ (s)	V_0 (m·s^{-1})	F_0 (N)	F_0 (N·kg^{-1})	P_{max} (W)	P_{max} (W·kg^{-1})	Drf (%)
Fornasier-Santos et al (21)	Catapult, 10	Intertrial	CV 1.47, % Δ 0.9, TEM −0.18	CV 5.74, % Δ 2.0, TEM 0.09	CV 1.63, % Δ 1.0, TEM 0.20		CV 5.64, % Δ −1.1, TEM −0.40			
Cormier et al (14)	Catapult	Intertrial			CV % 1.29, % Δ 0.15, TE 0.33, ICC 0.92		CV % 3.74, % Δ −0.77, TE 0.51, ICC 0.76		CV % 3.68, % Δ −0.66, TE 0.52, ICC 0.83	
Cormier et al (13)	Catapult, 10	Intrasystem	% Bias 0.23, TEM 0.02, ICC 0.98	% Bias 0.62, TEM 0.02 ICC 0.92	% Bias 0.31, TEM 0.03, ICC 0.97	%Bias −0.65, TEM 4.08, ICC 0.96		% Bias −0.31, TEM 6.82, ICC 0.98		% Bias −0.93, TEM 0.07, ICC 0.9
	STATSports, 10	Intrasystem	% Bias −0.39, TEM 0.02, ICC 0.98	% Bias 0.86, TEM 0.01, ICC 0.84	% Bias −0.37, TEM 0.02, ICC 0.97	% Bias −1.01, TEM 4.13, ICC 0.98		% Bias −1.37, TEM 5.65, ICC 0.99		% Bias −0.57, TEM 0.08, ICC 0.85
Clavel et al (11)	Catapult, 10	Intrasystem	CV % 0.5, TE 0.1, ICC 0.99, SWC 1.0%	CV % 2.0, TE 0.28, ICC 0.93, SWC 1.5%	CV % 0.6, TE 0.12, ICC 0.99, SWC 1.1%		CV % 1.8, TE 0.28, ICC 0.93, SWC 1.4%		CV % 1.4, TE 0.15, ICC 0.98, SWC 2.0%	
Huggins et al (32)	Polar Team Pro, 10	Intertrial	CV % 2.43, TE 0.18, SWC 0.06 m·s^{-1}							
Beato et al (8)	STATSports, 10 (Apex vs. Apex)	Intrasystem	CV % 0.12, ICC 0.99							
	STATSports, 10 (Viper vs. Viper)	Intersystem	CV % 0.20, ICC 0.97							
	STATSports, 10 (Apex vs. Viper)	Intermodel	CV % 0.15, ICC 0.98							
Barr et al (5)	SPI, 5	Intersystem	CV % 0.11, Δ 0.1, TEM 0.11, SWC 1.21%, r = 0.55							
Lacome et al (36)	V2 Sensor Everywhere, 16	Intrasystem—raw	CV % 0.5, TE 0.09, ICC 0.99, SWC 0.9%							
		Intrasystem—smooth	CV % 0.5, TE 0.09, ICC 0.99, SWC 0.9%							
Hoppe et al (31)	GPEXE, 18	Intrasystem			CV % 3.1		CV % 7.5		CV % 7.4	
	Catapult, 10	Intrasystem			CV % 3.3		CV % 20.9		CV % 18.8	
Nagarara et al (45)	GPEXE, 20	Intertrial	CV % 2.3	CV % 8.3	CV % 2.5		CV % 5.6		CV % 2.5	
	SPI, 5	Intertrial	CV % 5.1	CV % 17.5	CV % 6.0		CV % 19.2		CV % 15.8	
Akenhead et al (1)	Catapult, 10	Intrasystem—raw	CV % 15.6, TE 0.21, r = 0.98							
		Intrasystem—smooth	CV % 3.1, TE 0.08, r = 0.99							
Waldron et al (60)	SPI, 5	Intertrial	CV % 0.78							

*Sorted by year, CV = coefficient of variation, ICC = intraclass correlation coefficient, TE = typical error, TEM = typical error of measurement, SWC = smallest worthwhile change (reported from study), % Δ percent change between units (from same manufacturer), r = Pearson's correlation coefficient.

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Table 5
Moderator analysis results.*

Outcome		Subgroup random-effects model								
		k	SMD	95% CI		τ^2	I^2	*P* Value	Effect size	*p*
MSS (m·s^{-1})	**Sampling rate**									
	≥*10*	8	0.11	−0.04	0.26	<0.0001	9.40%	0.14	Small	0.001
	<*10*	1	0.75	0.38	1.11	—	—	<0.0001	Moderate	
	System									
	Catapult	4	0.02	−0.19	0.24	0	0.00%	0.83	Small	0.002
	StatSports	1	0.15	−0.20	0.50	—	—	0.40	Small	
	SensorEverywhere	2	0.59	0.18	0.99	0	0.00%	0.00	Moderate	
	GPEXE	1	−0.02	−0.33	0.29	—	—	0.88	Small	
	SPI	1	0.75	0.38	1.11	—	—	<0.0001	Large	
τ (s)	**Sampling rate**									
	≥*10*	6	−0.24	−0.71	0.24	0.3111	89.50%	0.33	Small	0.219
	<*10*	1	0.14	−0.22	0.49	—	—	0.82	Small	
	System									
	Catapult	4	−0.30	−0.68	0.08	0.0983	65.40%	0.12	Small	<0.001
	StatSports	1	0.68	0.32	1.04	—	—	0.06	Moderate	
	GPEXE	1	−0.86	−1.18	−0.53	—	—	0.02	Large	
	SPI	1	0.14	−0.22	0.49	—	—	0.71	Small	
V_0 (m·s^{-1})	**Sampling rate**									
	≥*10*	6	0.03	−0.12	0.19	0	0.00%	<0.0001	Small	0.001
	<*10*	1	0.73	0.37	1.10	—	—	0.68	Moderate	
	System									
	Catapult	4	0.02	−0.19	0.23	0	0.00%	0.86	Small	0.004
	StatSports	1	0.21	−0.14	0.56	—	—	0.24	Small	
	GPEXE	1	−0.08	−0.39	0.23	—	—	0.62	Small	
	SPI	1	0.73	0.37	1.10	—	—	<0.0001	Moderate	
F_0 (N·kg^{-1})	**Sampling rate**									
	≥*10*	6	0.21	−0.24	0.67	0.2814	89.10%	0.36	Small	0.150
	<*10*	1	−0.21	−0.56	0.14	—	—	0.71	Small	
	System									
	Catapult	4	0.23	−0.06	0.51	0.0347	41.40%	0.12	Small	<0.0001
	StatSports	1	−0.64	−0.99	−0.28	—	—	0.01	Moderate	
	GPEXE	1	0.95	0.62	1.27	—	—	0.00	Large	
	SPI	1	−0.21	−0.56	0.14	—	—	0.42	Small	
P_{max} (W·kg^{-1})	**Sampling rate**									
	≥*10*	4	0.24	−0.24	0.71	0.1952	85.60%	0.33	Small	0.670
	<*10*	1	0.11	−0.24	0.46	—	—	0.82	Small	
	System									
	Catapult	2	0.28	−0.19	0.74	0.067	58.10%	0.24	Small	<0.001
	StatSports	1	−0.35	−0.70	0.01	—	—	0.27	Small	
	GPEXE	1	0.69	0.38	1.01	—	—	0.02	Moderate	
	SPI	1	0.11	−0.24	0.46	—	—	0.73	Small	

*MSS = maximal sprint speed; *p*-diff = test for subgroup differences; SMDs = standardized mean differences (*d*); *k* = number of studies; CI = confidence interval.

(50). However, based on this review, it is not yet evident which filtering techniques are optimal for GNSS processing and more research needs to be performed to ensure consistency between comparisons.

Once appropriate data filtering is performed, other methodological decisions are required to prepare the data for sprint modeling such as determining the onset and end of sprint identification. When examining the reported onset of sprint detection and end of sprint detection, there were various methodologies used. These thresholds are often selected to avoid "noise" of the signal before the onset or inflection point of the sprint. In this review, onsets reported were as follows: when derived acceleration >0.1 m·s^{-2}, accelerometer acceleration >0 m·s^{-2}, synchronization with timing gates (3,32), velocity rising above a threshold deviation (e.g., +2 *SD*s) (36), velocity >0.3 m·s^{-1} (11), and velocity >0.5 m·s^{-1} with video correction (21). Concerning the onset, with current modeling practices, it is recommended that the lowest velocity or acceleration threshold be used based on the sampling rate of the system, although more robust techniques must be developed (to avoid signal noise and over or underestimations of profile metrics) and used consistently between studies to allow for result comparisons and to apply in the field. For end of sprint detection: set distance or integrated distance (13), velocity drop thresholds after peak velocity were achieved (e.g., −0.4 m·s^{-1}) (11,21), visually trimmed data, and synchronization with other technology (e.g., timing gates or software) (3,32) were used. The velocity modeling techniques of the studies included in this review were mostly monoexponential models (with or without time delay) (44,52). Considering that sprints are performed over short distances (up to 50 m), and knowing that this model implies a maximum asymptotic value where the highest point of the model should be near or at peak velocity of the athlete, using a cutoff shortly after peak velocity is achieved is the rationale choice when using this model with shorter distances. Accordingly, future research should assess how the different data-

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processing procedures affect sprint modeling metric outcomes with GNSS technology.

Various criterion devices were used in these studies (Table 3). However, not all studies used appropriate criterion devices for sprint profiling. For example, the use of timing gates for horizontal FV sprint profiling overestimates certain FV metrics associated with the acceleration phase of the sprint (i.e., τ, F_0, and P_{max}) (13,21). This is due to methodological considerations related to low sampling rates, sprint start limitations (e.g., fly-in distance to avoid the arm breaking the beam), beam system type (single vs. dual), and velocity modeling technique limitations (26). Therefore, using current sprint modeling practices, timing gates may not be suitable for use as a criterion measurement in validity studies, although highly practical as a tool in the field. These limitations may have caused the high percentage differences observed in some of the studies included in this review (Table 3, Figure 7). Recent evidence shows that using video to mitigate some sprint onset limitations can improve the agreement with radar (21); however, this still resulted in an overestimation in F_0 and as well this may not be an accessible approach to many sport science practitioners because of logistical considerations. Concerning other criterion technologies reported in this review, Fornasier-Santos et al. (21) showed that lasers and linear encoders have good agreement with radar for maximum velocity metrics, whereas force or acceleration metrics tend to display moderate to poor agreement. Radar provides practitioners with criterion reference that has been validated in previous research and may be superior due to the higher sampling rates and the ability to directly measure instantaneous velocity.

Distinction must be made between GPS and GNSS networks when reporting in sprint modeling studies because of differences in the sensors' (i.e., receiver) ability to acquire satellites. To accurately track position, a minimum of 4 satellites is required; thus, GNSS compatible systems allow for higher satellite network acquisition with some having the ability to acquire >10 satellites (9,11,13,56). Horizontal dilution of precision (hdop) shows the relationship between the organization of the satellites in relation to the receiver and positional measurement precision, where values can fall between 0 and 50, with values below 1 deemed beneficial (37). In this study, values from 5 Hz devices (older models) tended to exhibit hdop values >1 and newer (≥10 Hz) models tend to report values <1 (8,11,13,56,60). It is important to report the sensor networks, satellite acquisition, and positional precision values in GNSS or GPS sprint modeling research because sprint profiles may be sensitive to nuances in the quality of the signal due to environmental or location limitations (i.e., stadium structures, cloud cover, etc).

There are existing recommendations that sport scientists should evaluate when planning studies and disseminating findings from GNSS-derived outcomes (Malone et al. (37) and Ellen et al. (20)). This review adds to these recommendations by highlighting the need for standardization and transparent reporting of data processing to support best practice of sprint modeling. The first consideration is from the postmanufacturer processed data. Practitioners should evaluate whether the data are filtered or raw, and whether the acceleration data are derived or obtained from IMUs. It is recommended that the raw data are filtered or that sport scientists compare their own signal processing to that of the manufacturers to allow for replication of findings. As stated in previous research, the Doppler method is preferred over positional differentiation and information on signal acquisition and quality must be considered because this could affect the accuracy of the GNSS or GPS signal and consequently the sprint profiling. Second, when applying a sprint model to the data, it is important to consider that lower sampling rates or poor data processing procedures will lead to poor estimation of acceleration-based metrics; thus, it is recommended that GNSS with higher sampling rates are used and that the lowest possible thresholds for sprint onset be implemented. Data processing techniques such as sensor fusion may provide a solution to improve sampling and data fidelity, having already been proven in some studies (49). In addition, when modeling the sprint, the determination of sprint end needs to be considered in relation to the mathematical equation used

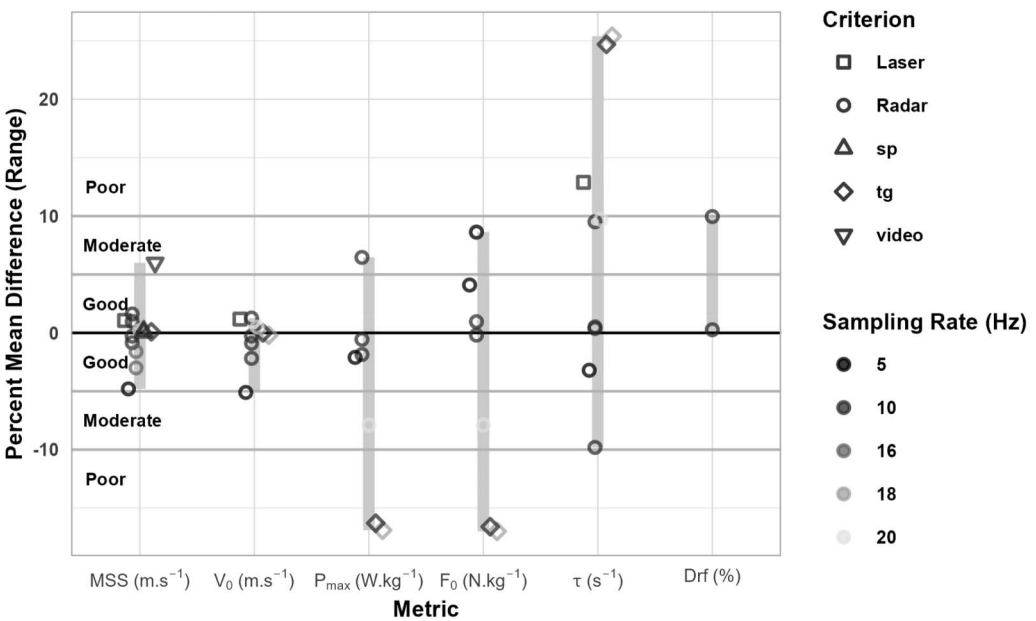

Figure 7. Range [min to max] percent mean difference values for GNSS radar or laser agreement reported with a priori percent agreement thresholds with interpretations. GNSS = global navigation satellite system; sp = sliding platform; tg = timing gates.

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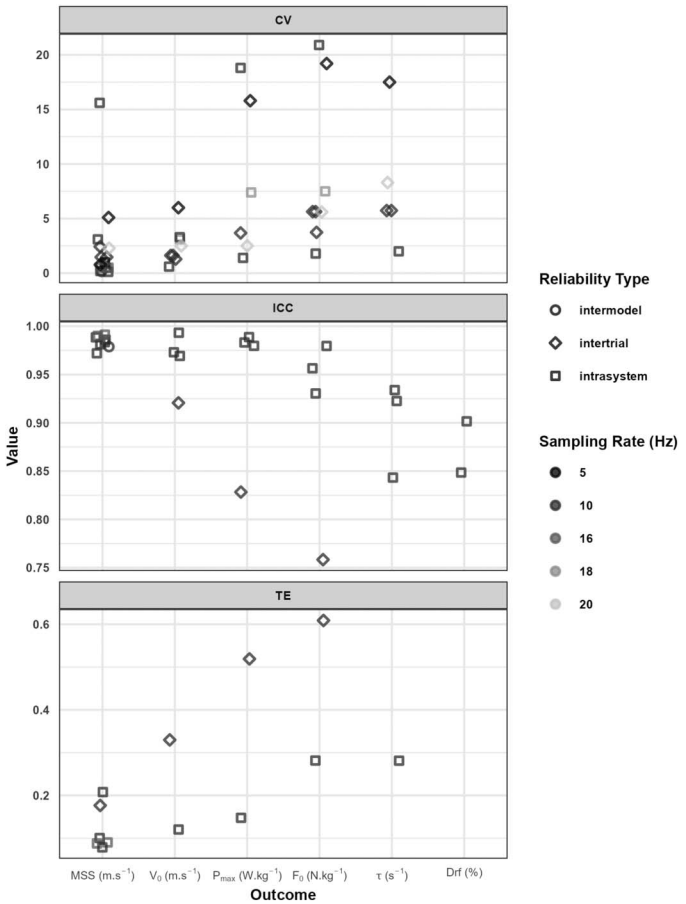

Figure 8. Intermodel, intertrial, and intrasystem reliability for each metric/study sorted by sampling rate (Hz).

for modeling. For example, data inputted into the monoexponential model should be trimmed at or shortly after the peak velocity is achieved. Other models (e.g., biexponential (41) or including fly-in velocities (59)) may necessitate different protocols concerning the onset and end of sprint detection. Therefore, studies to determine optimal sprint trimming parameters for differing collection protocols and techniques are required. Finally, it is important to report enough information for researchers and practitioners to replicate the methodology and compare findings across studies. Validity estimates (i.e., Bland-Altman, regression, etc.) with variability estimates (e.g., LOA, *SD*, or TE) as well as reliability statistics (e.g., CV, ICC, etc.) should be reported. Furthermore, information on athlete characteristics, sprint protocols, and specifications of the technology used (i.e., GNSS model, firmware, and software) and signal quality should be reported. Researchers are encouraged to share data-processing procedures through journal supplementary materials, online repositories, or have extremely detailed methodology outlined in their research. For example, 2 studies included in this review shared their coding protocols through supplementary materials or detailed protocols in their methodology (11,13). The application of the recommendations made in this review will allow for accelerated improvement of current sprint modeling practices in future research.

As it has been shown that it is possible to derive valid and reliable FV profiles from GNSS data given that appropriate methodology is conducted by sport science practitioners, potential for application in the field is promising. Studies have suggested the possibility to integrate multiple standardized maximal sprint efforts (e.g., 2 × 40–50 m sprints) to provide systematic information on athlete sprint acceleration capabilities (11,36). These studies have included coding script to efficiently process the data included in supplementary materials. Hicks et al. (28) provided practical recommendations to improve mechanical effectiveness of force application and recently introduced a framework to profile and individualize programs to address force and velocity components (27). For example, an approach that could be taken is plotting athletes' V_0 on the *x*-axis and F_0 on the *y*-axis with a scatterplot divided into 4 quadrants by the medians of each variable, then identifying deficits to target with different training modalities (e.g., resisted sprinting or resistance training) (27) or identify strengths to maximize. In team sports with large rosters, it may be beneficial to group athletes based on their FV characteristics to inform training programs. These methods could especially be beneficial for sports in which long linear or semilinear sprints are frequently performed in training and competition such as soccer, rugby codes (e.g., rugby sevens), football (American and Australian rules), and so on. The actual integration of sprints into normal training has now been demonstrated in 2 cases, both displaying valid and reliable results (10,14).

This review outlines the currently available literature on linear sprint modeling with GPS or GNSSs, identifying gaps and opportunities for improvement. Primarily, sport science practitioners are encouraged to open dialog with sport technology companies to better understand GNSS units' proprietary signal processing algorithms. If this information cannot be shared, sport scientists should consider the postmanufacturer processing, sprint

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model processing, and reporting of sprint protocol and specification of the GNSS technology used (20,37). This will allow for better comparisons of the findings across studies and improvements in future research.

Practical Applications

Although there are some methodological challenges to sprint modeling using GNSS technology, validity and reliability studies suggest that sprint modeling can be performed and show promise for sprint monitoring in the field. The key consideration is that the most up-to-date modeling practices should be used consistently with similar sprint protocols because the methodology will affect sprint profiling outcomes. As suggested in previous research (11,13,35,36), the integration of maximal standardized linear sprint efforts can be integrated into training sessions with team-sports and individual sport athletes. Systematic sprint modeling could allow performance teams to monitor fatigue and readiness, better-informing periodization and training stimulus prescription. This has important implications for concepts like "invisible monitoring" (i.e., testing without testing) (14,42), systematic testing, fatigue monitoring, rehabilitation, performance, and training design.

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